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Vortex86DX2 (A9125B)
Power Consumption

2012/01/05

Version 1.1

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Revision History

Rev.	Date	Editor	Executor	History
REV 1.0	2011/10/28	Yanmin	Yanmin	1. Original version
REV 1.1	2012/01/05	Yanmin	Yanmin	1. EVB Status 加入測試時的 CPU/DRAM clock 資訊. 2. Test Status 加入測試時的詳細資訊. 3. 修正 RTC standby current 資訊.

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1 Test Goal

- Measure the A9125B BIN3 IC leakage
- Test A9125B BIN3 IC power consumption

2 EVB Status

A9125B EVB 114 (BIN3 IC)

Dram 1G (Nanya NT5TU128M8GE-AC*8)

CPU clock: 1GHz

Dram clock: 400MHz

3 Test status

1. Vcore dynamic current：7-Zip in Windows XP
   Note: 測試時將電壓維持在 1.0V, 四種電源中以 Vcore 部份最容易受到溫度影響導致電壓不正確.
2. V1v8 dynamic current：Memtest 211 in DOS
   Note: 測試時將電壓維持在 1.8V
3. V1v2 dynamic current：HDD test in Windows XP
   Note: 測試時將電壓維持在 1.2V
4. V3v3 dynamic current：USB Disk+HDD+MAC test in Windows XP
   Note: 測試時將電壓維持在 3.3V

4 Summary

- Leakage
  1. A9125B RTC standby current
     - input voltage 3v standby current 介於 2.1uA ~ 2.3uA
- Power consumption
  1. A9125B BIN3 IC Power Consumption Max Data ~ 6.451 W at 70℃
     - Vcore：Max Data ~ 3.452 W at 70℃
     - V1v2：Max Data ~ 0.439 W at 70℃
     - V1v8：Max Data ~ 1.438 W at 70℃
     - V3v3：Max Data ~ 1.122 W at 70℃

  Note: Power Source
  - Vcore：CPU core power, DAC digital power
  - V1v2：SB core power, Differential PAD, Internal Ethernet PHY digital logic cell, PCI-E, SATA PHY, USB
  - V1v8：SB analog power, DDR, DRAM PLL, CPU PLL, Internal Ethernet PHY, DAC, VGA PLL, Temperature Sensor, Internal PLL
  - V3v3：I/O PAD, PCI-E, SATA PHY, USB, ADC

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5 Leakage

5.1 A9125B Leakage

		Vcore (mA)	V1v2 (mA)	V1v8 (mA)	V3v3 (mA)
A9125B_114 (BIN3)	35℃	1232	70	366	127
	60℃	1860	72	366	128
	70℃	2170	72	366	129
	85℃	2550	73	366	129

5.2 A9125B RTC standby current

RTC capaci discharge finish

Input Voltage	2V	2.5V	3V	3.3V
A9125B_114 (BIN3)	0.8 uA	1.5 uA	2.2 uA	3.0 uA

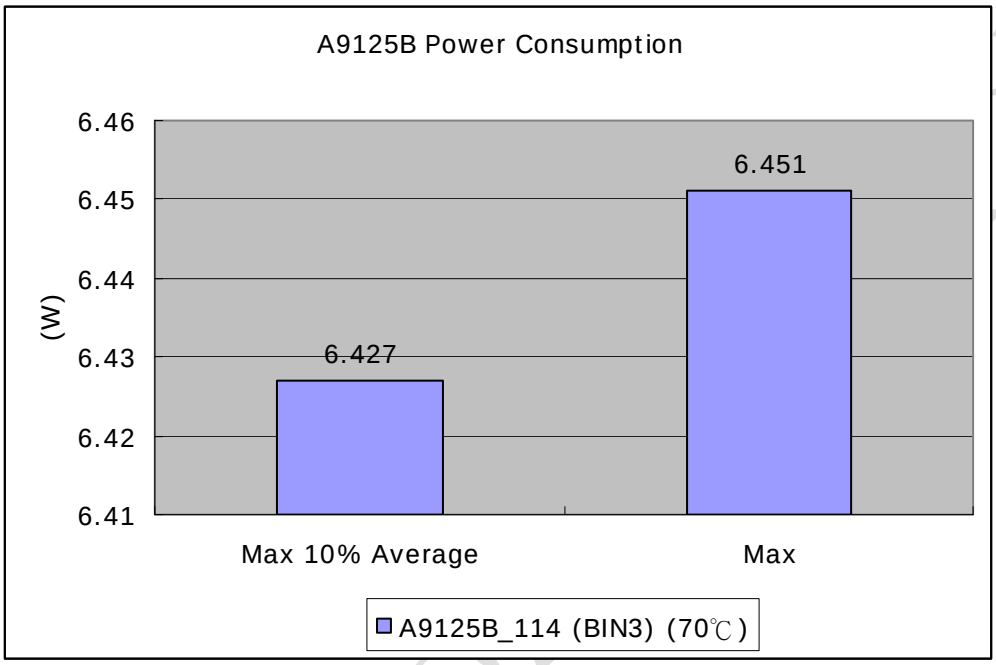
Input Voltage	3V
A9125B_114 (BIN3)	2.2 uA
A9125B_117 (BIN1)	2.1 uA
A9125B_120 (BIN2)	2.3 uA

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6 Power Consumption

6.1.1 Total power consumption

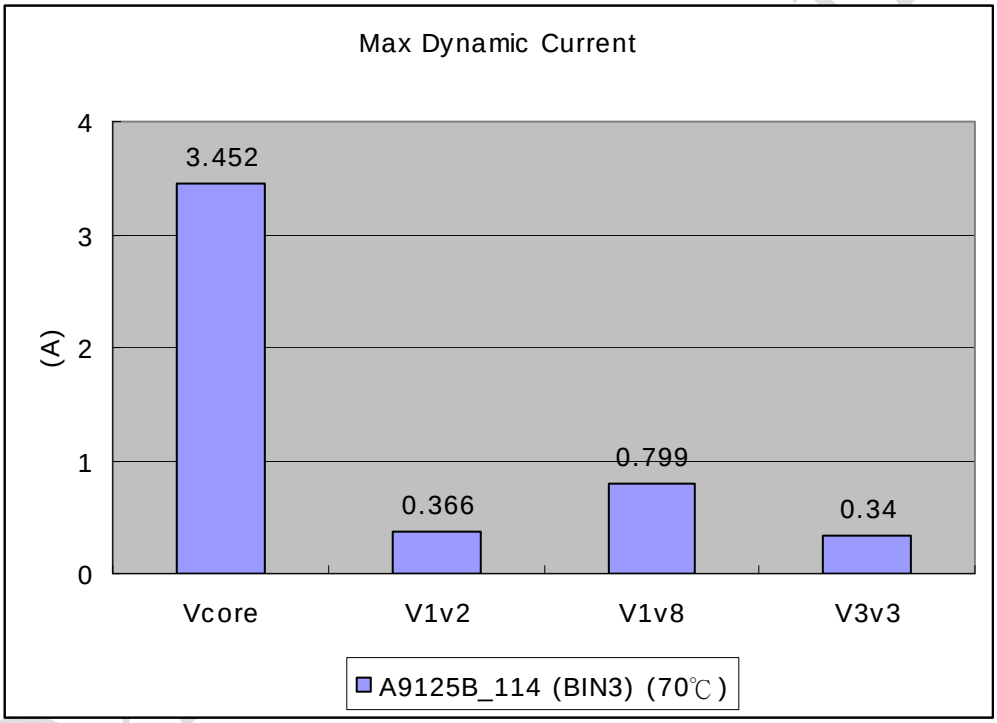
Total Power Consumption	Max 10% Average (W)	Max (W)
A9125B_114 (BIN3) (70℃)	6.427	6.451



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6.1.2 Max dynamic current

Max Dynamic Current	A9125B_114 (BIN3) (70℃) (A)
Vcore	3.452
V1v2	0.366
V1v8	0.799
V3v3	0.340



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6.2 A9125B Dynamic Current

6.2.1 Vcore

V1v0 Dynamic Current	Max 10% Average (A)	Max (A)
A9125B_114 (BIN3) (70℃)	3.443	3.452

6.2.2 V1v2

V1v2 Dynamic Current	Max 10% Average (A)	Max (A)
A9125B_114 (BIN3)	0.364	0.366

6.2.3 V1v8

V1v8 Dynamic Current	Max 10% Average (A)	Max (A)
A9125B_114 (BIN3)	0.797	0.799

6.2.4 VDDR

VDDR Dynamic Current	Max 10% Average (A)	Max (A)
A9125B_114 (BIN3)	0.644	0.648

6.2.5 V3v3

V3v3 Dynamic Current	Max 10% Average (A)	Max (A)
A9125B_114 (BIN3)	0.337	0.34

7 Power Suggestion

DMP suggestion to use 6 Amp Power regulator for Vcore to ensure the stability in high temperature environment.

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